

Jaisidh Singh

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Education

2024–present **MSc in Machine Learning**, *Eberhard-Karls Universität Tübingen*, Tübingen.
(expected 2027) ○ German GPA: 1.52/5.0 (1.0 is best)

2020–2024 **B.Tech in AI & Data Science**, *Indian Institute of Technology*, Jodhpur.
○ Thesis: negation-aware compositional reasoning in vision-language models.

Awards & fellowships: Konrad Zuse School (ELIZA) Fellowship for Master's Students in Germany 2024–2026.

Work Experience

Oct 2026 – **AI Research Intern at Aleph Alpha**, *With Dr. Yasser Jadidi & Dr. Leonard Salewski*.
present ○ Working on data generation for SFT.

Oct 2025 – **SWE Assistant at Tübingen AI Center**, *With Prof. Peter Gehler*.
May 2026 ○ Developed & **deployed real-time interactive ML systems** for stylization at Stadtmuseum Tübingen.
○ Created & **delivered a hands-on workshop** on agent orchestration at **GCPR 2025**.

Jan 2025 – **ML Research Assistant at ELLIS Institute Tübingen**, *With Dr. Antonio Orvieto*.
May 2025 ○ **Reduced VLM training cost by 90%** for modality stitching by using a hypernetwork-based framework.
Published: EMNLP 2025 Main Conference.

Summer 2022 & 2023 **AI Research Intern at Bosch Research India**, *With Dr. Amit A. Kale & Sonam Singh*.
○ **Developed an interpretable failure-discovery system** for semantic segmentation models on industry-scale autonomous driving data **achieving 95% precision**. **Technical report: arXiv:2411.10845**.
○ **Created multi-modal image retrieval pipelines** analyzing weather effects on AV cameras.

Research Experience

Apr 2026 – **Thesis Research at OpenEuroLLM**, *With Dr. Aaron Klein & Dr. Antonio Orvieto*.
present ○ **Improving data efficiency ($\approx 30\%$)** in LLM pretraining by scaling hybrid-attention architectures.
○ Collaborated on **novel linear attention architecture research**, adding LLM experiments.
Published: NeurIPs 2026 NeuReps Workshop.

Oct 2025 – **Graduate Research at ELLIS Institute Tübingen**, *With Dr. Antonio Orvieto*.
Jan 2026 ○ **Investigated grokking dynamics in transformers**, showing CE+AdamW does not admit kernel regime & layer-norm modulates grokking speed. **Published: ICLR 2026 Trustworthy AI Workshop.**

Winter 2022 & 2023 **Undergraduate Research at Trusted AI Lab IITJ**, *With Prof. Mayank Vatsa, Prof. Richa Singh*.
○ Designed a compositional negation benchmark & fine-tuned CLIP on custom data, **achieving SOTA accuracy on negation understanding**. **Published: WACV 2025 (60+ citations)**.
○ Systematically analyzed StyleGAN2 latent space to profile identity leakage. **Published: WACV 2024.**

Selected Publications

2026 **Complex KDA: Understanding and Enhancing the Expressivity of Kimi Delta Attention**, *arXiv preprint*, Julien Siems*, Riccardo Grazi*, Korbinian Pöppel*, Jaisidh Singh, Arber Zela, Timur Carstensen, Jenia Jitsev, Frank Hutter, Volkan Cevher, Antonio Orvieto, Aaron Klein.

2025 **(Almost) Free Modality Stitching of Foundation Models**, *EMNLP 2025, ICLR-W 2025*, Jaisidh Singh, Diganta Misra, Boris Knyazev, Antonio Orvieto.

(Refer to my Google Scholar profile for a complete list of publications. * indicates equal contribution.)

Technical Skills

- **Machine learning:** PyTorch, JAX, FLAX, Haiku, Optax, Huggingface
- **Distributed training & HPC:** Slurm, Megatron-LM, Sharding/distributed training
- **Programming languages:** Python, C++, Bash, JavaScript, Dart